ELEC2302/9302 Signals and Systems

Bonus quiz 3 — version A

Name: _____ unikey: _____

Total marks 17.

You are eligible for this bonus quiz if your highest Quiz 3 mark so far is $\geq 85\%$ (13/15). If you do not meet this requirement you will not receive bonus marks.

Show all working. Use the large boxes for **working** and the small labelled boxes for **final answers**. The formula sheets are attached at the back.

1. A system has transfer function

$$G(s) = \frac{1 - e^{-sT}}{sT}, \qquad T > 0.$$

[7 marks]

(a) Find the impulse response $h(t)$. **[2 marks]**

$h(t) =$

(b) Find the step response. **[3 marks]**

(c) Is the system BIBO stable? Justify your answer. **[2 marks]**

2. The Chebyshev polynomials are $T_n(x) = \cos(n \arccos x)$ for $|x| \le 1$, with $T_0(x) = 1$, $T_1(x) = x$ and $T_{n+1}(x) = 2x\,T_n(x) - T_{n-1}(x)$ for all x. A Chebyshev lowpass filter of order n with cutoff ω_c has

$$|G(j\omega)|^2 = \frac{1}{1 + \varepsilon^2\, T_n^2(\omega/\omega_c)}, \qquad \varepsilon > 0.$$

[10 marks]

(a) Show that for $\omega \le \omega_c$ the gain lies between $\dfrac{1}{\sqrt{1+\varepsilon^2}}$ and 1. **[2 marks]**

(b) The passband ripple is the ratio of the largest to the smallest gain in the passband, in dB. Show that it is $10 \log_{10}(1+\varepsilon^2)$ dB, and find ε for a ripple of 1 dB. **[2 marks]**

$\varepsilon =$

(c) Use the recurrence to find $T_2(2)$, $T_3(2)$, $T_4(2)$ and $T_5(2)$. **[2 marks]**

$T_2(2) =$

$T_3(2) =$

$T_4(2) =$

$T_5(2) =$

(d) Write the gain at $\omega = 2\omega_c$ in dB, in terms of ε and $T_n(2)$. With the ripple of 1 dB, find the smallest order n for which the gain at $2\omega_c$ is at most -40 dB. **[4 marks]**

Gain at $2\omega_c =$ dB

$n =$

ELEC2302/9302 Signals and Systems — Quiz 3 formula sheet

Laplace transform

$$U(s) = \mathcal{L}\{u\}(s) = \int_{0^-}^{\infty} u(t)\, e^{-st}\, \mathrm{d}t, \qquad Y(s) = G(s)\, U(s), \qquad G(s) = \mathcal{L}\{h\}(s).$$

$u(t)$	$U(s)$	$u(t)$	$U(s)$
$\delta(t)$	1	$e^{-at}H(t)$	$\dfrac{1}{s+a}$
$\delta(t-t_0),\ t_0 \geq 0$	e^{-st_0}	$t\,e^{-at}H(t)$	$\dfrac{1}{(s+a)^2}$
$H(t)$	$\dfrac{1}{s}$	$t^n e^{-at}H(t)$	$\dfrac{n!}{(s+a)^{n+1}}$
$t\,H(t)$	$\dfrac{1}{s^2}$	$e^{-at}\cos(\omega_0 t)H(t)$	$\dfrac{s+a}{(s+a)^2+\omega_0^2}$
$t^n H(t)$	$\dfrac{n!}{s^{n+1}}$	$e^{-at}\sin(\omega_0 t)H(t)$	$\dfrac{\omega_0}{(s+a)^2+\omega_0^2}$
$\cos(\omega_0 t)H(t)$	$\dfrac{s}{s^2+\omega_0^2}$	$\sin(\omega_0 t)H(t)$	$\dfrac{\omega_0}{s^2+\omega_0^2}$

Properties of the Laplace transform

	$u(t)$	$U(s)$		$u(t)$	$U(s)$
linearity	$\alpha u_1(t) + \beta u_2(t)$	$\alpha U_1(s) + \beta U_2(s)$	differentiation	$\dfrac{\mathrm{d}}{\mathrm{d}t}u(t)$	$s\,U(s) - u(0^-)$
time shift	$u(t-t_0)H(t-t_0),\ t_0 \geq 0$	$e^{-st_0}\,U(s)$	second derivative	$\dfrac{\mathrm{d}^2}{\mathrm{d}t^2}u(t)$	$s^2U(s) - s\,u(0^-) - \dot{u}(0^-)$
s-shift	$e^{-at}u(t)$	$U(s+a)$	integration	$\displaystyle\int_{0^-}^{t} u(\varsigma)\,\mathrm{d}\varsigma$	$\dfrac{U(s)}{s}$
time scaling	$u(at),\ a>0$	$\dfrac{1}{a}\,U\!\left(\dfrac{s}{a}\right)$	convolution	$(h * u)(t)$	$H(s)\,U(s)$
multiply by t	$t\,u(t)$	$-\dfrac{\mathrm{d}}{\mathrm{d}s}U(s)$	modulation	$u(t)\cos(\omega_0 t)$	$\frac{1}{2}U(s-j\omega_0) + \frac{1}{2}U(s+j\omega_0)$

Initial value theorem: $u(0^+) = \lim\limits_{s\to\infty} s\,U(s),$ **Final value theorem:** $\lim\limits_{t\to\infty} u(t) = \lim\limits_{s\to 0} s\,U(s).$

Butterworth filters

$$|G(j\omega)|^2 = \frac{1}{1+(\omega/\omega_c)^{2n}}, \qquad n=1:\ \frac{\omega_c}{s+\omega_c}, \qquad n=2:\ \frac{\omega_c^2}{s^2+\sqrt{2}\,\omega_c s+\omega_c^2}, \qquad n=3:\ \frac{\omega_c^3}{(s+\omega_c)(s^2+\omega_c s+\omega_c^2)}.$$

$$n \text{ even: } G(s) = \prod_{k=1}^{n/2} \frac{\omega_c^2}{s^2+2\zeta_k\omega_c s+\omega_c^2}, \qquad n \text{ odd: } G(s) = \frac{\omega_c}{s+\omega_c}\prod_{k=1}^{(n-1)/2} \frac{\omega_c^2}{s^2+2\zeta_k\omega_c s+\omega_c^2}, \qquad \zeta_k = \sin\!\left(\frac{(2k-1)\pi}{2n}\right).$$

Modulation

AM	$y(t) = u(t)\cos(\omega_c t), \qquad Y(j\omega) = \frac{1}{2}U\big(j(\omega-\omega_c)\big) + \frac{1}{2}U\big(j(\omega+\omega_c)\big)$
Demodulation	$z(t) = y(t)\cos(\omega_c t) = \frac{1}{2}u(t) + \frac{1}{2}u(t)\cos(2\omega_c t)$
FM	$y(t) = \cos\!\Big(\omega_c t + k\displaystyle\int_{-\infty}^{t} u(\tau)\,\mathrm{d}\tau\Big)$
PM	$y(t) = \cos\big(\omega_c t + k\,u(t)\big)$

First and second order systems

First order, $G(s) = \dfrac{K}{\tau s+1}$: the step response reaches 63% of its final value at $t=\tau$, rise time (10% to 90%) $\approx 2.2\tau$, settling time (to within 5%) $\approx 3\tau$.

Second order, $G(s) = \dfrac{K\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$, $0<\zeta<1$:

$$\text{settling time (5\%)} \approx \frac{3}{\zeta\omega_n}, \qquad \text{resonant peak } \frac{|G(j\omega_r)|}{|K|} = \frac{1}{2\zeta\sqrt{1-\zeta^2}} \text{ at } \omega_r = \omega_n\sqrt{1-2\zeta^2}, \qquad \zeta < \tfrac{1}{\sqrt{2}}.$$

ζ	0.1	0.15	0.2	0.25	0.3	0.4	0.5	0.6	0.707
overshoot M_p	73%	62%	53%	44%	37%	25%	16%	9.5%	4.3%
resonant peak $\lvert G(j\omega_r)\rvert/\lvert K\rvert$	5.03	3.37	2.55	2.07	1.75	1.36	1.15	1.04	1.00
resonant peak, dB above DC	14.0	10.6	8.1	6.3	4.8	2.7	1.2	0.4	0

ELEC2302/9302 Signals and Systems — Quiz 1 formula sheet

Complex exponentials

$$e^{j\theta} = \cos\theta + j\sin\theta$$

A complex number in Cartesian and polar form:

$$z = x + jy = r\,e^{j\theta}, \qquad r = |z| = \sqrt{x^2+y^2}, \qquad \theta = \arg z.$$

Inner product, energy and power

$$\langle u, y\rangle := \int_{-\infty}^{\infty} u(t)\,\overline{y}(t)\,\mathrm{d}t$$

$$E(u) = \int_{-\infty}^{\infty} |u(t)|^2\,\mathrm{d}t, \qquad P(u) = \lim_{W\to\infty} \frac{1}{2W}\int_{-W}^{W} |u(t)|^2\,\mathrm{d}t, \qquad u_{\mathrm{RMS}} = \sqrt{P(u)}.$$

If u is periodic with period T, then

$$\langle u, y\rangle_P = \frac{1}{T}\int_0^T u(t)\,\overline{y}(t)\,\mathrm{d}t, \qquad P(u) = \langle u, u\rangle_P = \frac{1}{T}\int_0^T |u(t)|^2\,\mathrm{d}t.$$

Standard signals

Unit step (Heaviside).

$$H(t) = \int_{-\infty}^{t} \delta(\varsigma)\,\mathrm{d}\varsigma = \begin{cases} 0, & t < 0 \\ 1, & t \geq 0 \end{cases}$$

Pulse of width T.

$$p_T(t) = \begin{cases} 1, & |t| \leq T/2 \\ 0, & \text{otherwise} \end{cases}$$

$$p_T(t) = H\left(t + \tfrac{T}{2}\right) - H\left(t - \tfrac{T}{2}\right), \qquad \frac{\mathrm{d}}{\mathrm{d}t}H(t) = \delta(t).$$

The Dirac delta

Sifting property.

$$\int_{-\infty}^{\infty} \delta(t-\tau)\,y(t)\,\mathrm{d}t = y(\tau).$$

Scaling.

$$\delta(\alpha t) = \frac{1}{|\alpha|}\,\delta(t).$$

Convolution

$$(h * u)(t) := \int_{-\infty}^{\infty} h(t-\tau)\,u(\tau)\,\mathrm{d}\tau = (u * h)(t)$$

$$(u * \delta)(t) = u(t), \qquad \big(u * \delta(t-T)\big) = u(t-T), \qquad h_1 * (h_2 * h_3) = (h_1 * h_2) * h_3.$$

Impedances

$$Z_R = R, \qquad Z_L = j\omega L, \qquad Z_C = \frac{1}{j\omega C}.$$